The Web of Finance

By replacing platform-dependent interoperability with portable, platform-independent, and universally verifiable proofs, the financial industry can evolve into an open, interoperable “web of finance,” where value moves as freely as information on the internet.

DLT is a design error:

- DLT uses cryptography to secure data inside a closed platform (a Ledger),
- From an architectural standpoint, a DLT Ledger is a "walled garden" by design. It relies on a **Global State Machine**—forcing every participant to synchronise with a single platform to prove a transaction.

DLT simply adds a layer of cryptography to a **monolithic, centralised execution platform.** Just like the mainframes of the past, DLT requires:

- **A Single Platform:** Every participant must be "on" the same chain/network to interact.
- **Forced Co-location:** Logic and data are bound to the specific ledger’s rules and consensus engine.
- **Locked-in Trust:** Trust is anchored to the platform’s health and its specific validator set, rather than the data itself.

By forcing all users into a single "global state," we do not decentralise finance; we’ve just built a more expensive, cryptographically-signed mainframe.
There is no disruption or innovation, simply a reengineering of existing platforms: DLT is used to deploy new monolithic platforms. Just as we did with mainframes, we need additional platforms to enable interoperability among monoliths.

The Shift: Web of Finance - W3C Verifiable Credentials (VCs)
Cryptography’s true power is its ability to **decouple trust from the environment.** The goal of cryptography should be to facilitate the exchange of trust records between **autonomous agents**—not to secure a single, closed platform. Using cryptography to build a walled-garden ledger is a fundamental waste of the technology's potential.

The true "Web 3.0" for finance isn't a ledger; it’s a **cryptographically-enabled network of verifiable data.** By moving to **W3C Verifiable Credentials (VCs)**, we leverage the original promise of the internet:

1. **Autonomous Agents:** Institutions and individuals manage their own data and keys, acting as independent nodes.

2. **Platform Independence:** A VC issued on Platform A can be verified by a user on Platform B without any technical bridge or shared state.
3. **Peer-to-Peer Trust:** Trust is embedded in the **data packet** itself, not the platform it sits on.

The source of trust is no longer the platform or ledger but the credential issuer, who attests to and guarantees the claims.

The real innovation isn't a "shared ledger"; it's the ability to exchange **unforgeable, portable data.**
VCs use cryptography to facilitate the **exchange of trust records** across any protocol (HTTPS, DIDComm, etc.).

- **Zero Platform Dependency:** You don't need to "join a network" to verify a VC. You only need the issuer's public key.
- **Information Symmetry:** All parties can share cryptographically-proven data, ensuring trust without the need for a shared "middleman" ledger.
- **Granular Privacy:** VCs natively support selective disclosure. You can prove you are "Over 21" or "AML Compliant" without revealing the underlying PII or ledger history.

From Platforms to Protocols
To build a truly interoperable financial system, we must stop building "platforms" and start building **trust protocols.** Interoperability shouldn't require a "bridge" between two mainframes; it should be as simple as an autonomous agent presenting a signed credential to another. One is a closed loop; the other is a web.

The emergence of a "web of finance" extends the principles of the web—open communication, interoperability, and direct interaction—to the exchange of value.

Just as the web transformed communications by removing intermediaries and enabling direct, global information exchange, a proof-based financial system enables direct, global value exchange between participants.

This shift redefines trust relationships: instead of relying on layered intermediaries, participants can rely on verifiable proofs and shared standards. The result is a profound simplification of financial interactions—reducing friction, lowering costs, and enabling new forms of business collaboration. As the web unlocked new economic models in information exchange, the web of finance has the potential to unlock equally transformative innovation in financial services and commerce.

Example: Payment

All payment systems—cards, bank transfers, and DLT—rely on three fundamental assurances:

- **Proof of Ability** establishes that the parties can transact
- **Proof of Capability** ensures that the payer can afford the transaction
- **Proof of Intent** commits the payer to the transaction

This model provides a **technology-neutral abstraction** that explains all existing payment systems and enables **seamless cross-domain payments**.

- In all existing payment systems, proofs are issued and ultimately validated by their originating issuers: a Bank, a Card system, or a DLT Ledger.
- Interoperability between different issuers is achieved through platforms—such as card networks, banking settlement systems, or DLT bridges—that enable these proofs to be transported, interpreted, and acted upon across domains.

Card payment systems are platforms created to enable peer-to-peer interactions (Client-Merchant) that were not possible on banks' platforms.

DLT is working on the same line: peer-to-peer payment is only possible through the same ledger. If the Payer and the Payer are on different ledgers, another platform (a bridge) is needed to enable interoperability. Considering the fragmentation of DLT solutions (CBDC, Stablecoins, ...), the number of bridging platforms will grow exponentially.

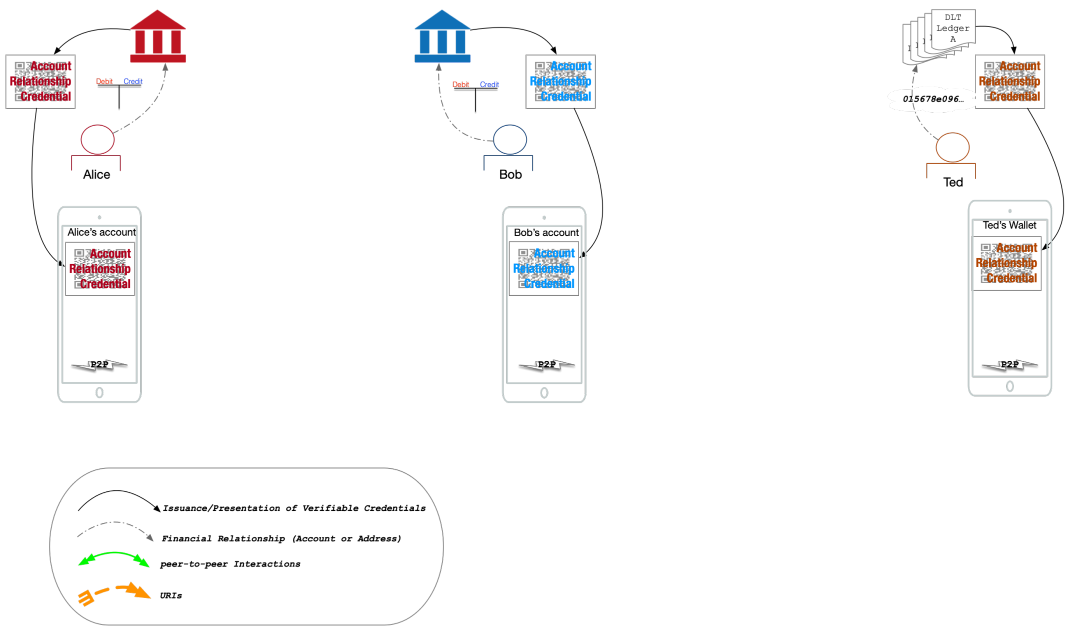
Digital transformation enables a fundamental shift:
Transitioning from platform-mediated interoperability to proof-based interoperability will significantly revolutionise the financial sector.

- **Proofs of Ability** become portable credentials (“financial passports”), issued by the account (Bank) or address servicer (a Ledger)
- **Proofs of Capability** are standardised and verifiable, issued by the Party controlling the money (a Bank or a Ledger) that is also blocking the money.
- **Proofs of intent** are issued by the Payer, embedding proofs of ability and capability.
- Proofs are:
  - platform-independent
  - portable across domains
  - verifiable by any participant
- **Proofs provide information symmetry**: They contain the certified information necessary to enable execution, such as the account

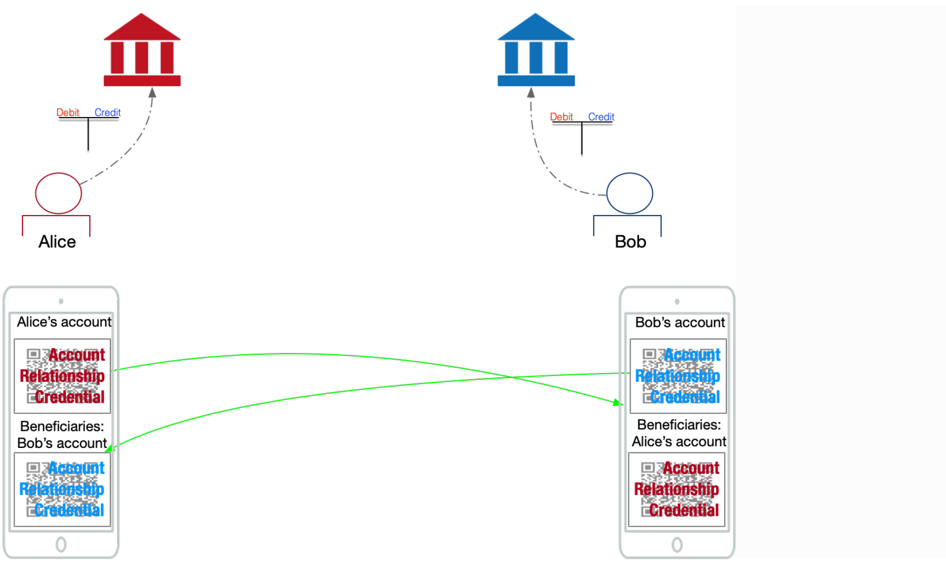
number, the name of the account owner, and the list of supported domains (settlement services). For example, a bank could support SEPA Inst, SWIFT, and a bridge to a DLT ledger.

Under this approach:

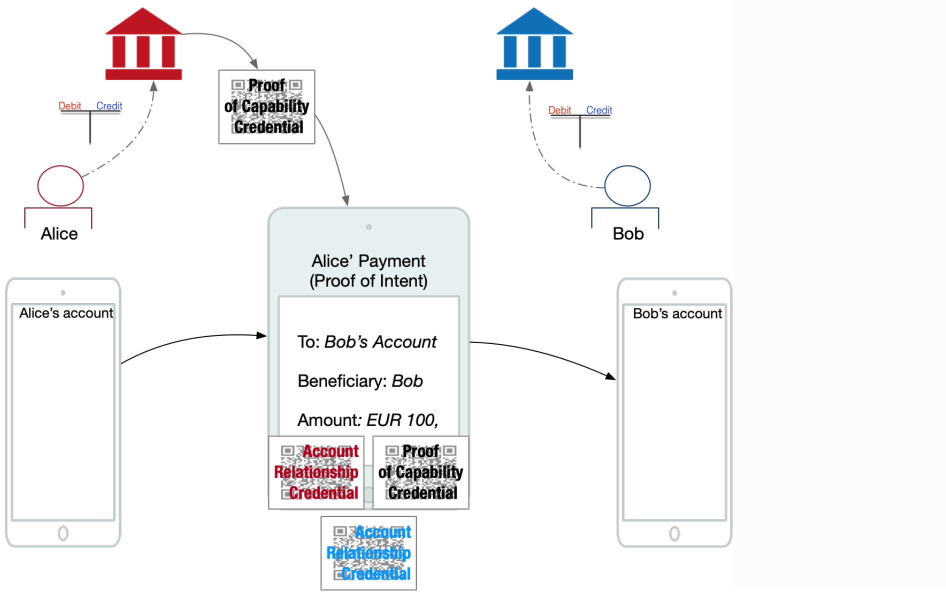
- Authorised parties (Banks, Ledgers, Card systems) issued portable proofs of ability: Account Verifiable Credentials (ARC)



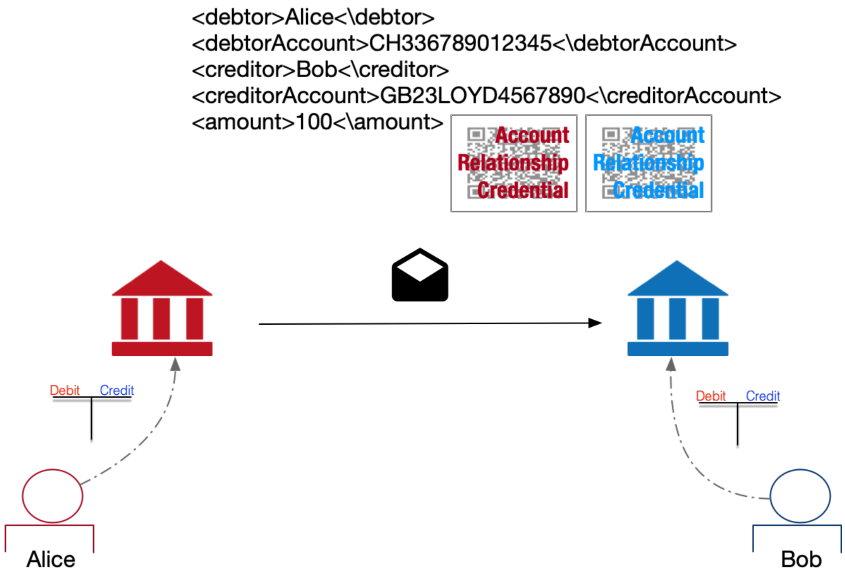
- Payers and payees exchange proofs directly:



- Proof of capability is issued by the Payer's agent (Account Servicing Bank, Issuing Bank or Ledger)
- The Proof of Intent is issued by the Payer.



- Compatibility is determined by the list of services supported that is given in the financial passport:
  - shared execution domains, or
  - reachable paths across domains
- Settlement happens between the financial agents, based on the information provided in the ARCs.



- Platforms (networks, rails, bridges):
  - are no longer required as trust intermediaries
  - become **execution and routing layers**

Interoperability is no longer pre-built—it is dynamically established through proof of compatibility.

The benefits:

- A unified user experience is provided for online, credit transfer, or counter payment methods, irrespective of platform or currency dependence.
- The peer-to-peer protocol facilitates the sharing of financial coordinates, payment proof of intent, and may also enable the sharing of payment obligations (such as invoices or sales tickets) as Verifiable Credentials.
- Eliminate frictions by enabling each custodian to verify the identities of both the Payer and Payee. This approach obviates the necessity for "verification of payee" infrastructure or "Travel Rule" off-chain exchanges, while concurrently safeguarding privacy.
- Efficiency: The flows of information are restricted to the Parties involved (Payer, Payer's Agent, Payee, and Payee's Agent).
- Simplified trust, acountability and verifiability
- By establishing links and automating payments and payment obligations, verification processes, traceability, and accountability are rendered more straightforward.

Cross-Domain Payments as a Native Capability

With portable and verifiable proofs:

- Payments can occur across domains **whenever a valid path exists.** It essentially depends on banks, ledgers, and card networks developing interoperability, but it has a limited impact on users' experience.
- Bridges and interoperability mechanisms become:
  - infrastructure-level capabilities
  - transparent to end users

End users:

- present their ability
- verify counterparties
- commit to payments

They do **not**:

- manage domains
- select routing
- interact with bridges

Payments become domain-agnostic at the user level and domain-resolved at the infrastructure level.

#SystemArchitecture #DLT #Web3 #VerifiableCredentials #Cryptography #FinTech #W3C #DigitalTransformation